Advancing Social Insight with Large Language Models: Unpacking a Geopolitical Conflict at Scale

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How do people engage with prolonged geopolitical conflicts on social media?

Real-time construction of narratives, meaning
(Anden-Papadopoulos, 2013; Zelizer, 2010)

Implications for media coverage, humanitarian response, political advocacy
(Zeitzoff, 2018)

Reliance by governments, NGOs, media organizations
(Ghodes & Steinert-Threlkeld, 2025)

Initially intense, heightened distress

(Garcia & Rime, 2019; Guntuku et al., 2020)

Linguistic indicators of emotion

(Boyd & Schwartz, 2021)

1.2M+ tweets, Russia-Ukraine War, Feb-Dec 2022

Language Inquiry & Word Count

(Boyd et al., 2022)

More – valenced language, less + language during early stages

People are less upset

OR

Conversation shifted from shock to sanctions and energy policy

ADAPTATION OR TOPIC DRIFT?

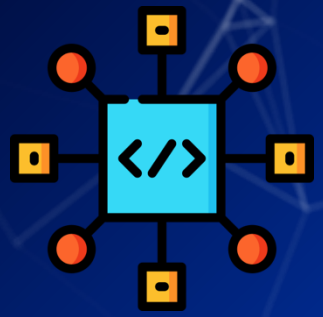
Tweet 1: These war crimes are disgusting and evil. The destruction and suffering are shameful. Russia must be held accountable.

Tweet 2: Calling this ‘war crimes’ is disgusting propaganda. The outrage is ridiculous and shameful. NATO’s reckless expansion created this catastrophe.

SENTIMENT, NOT NECESSARILY ENGAGEMENT

CONTEXT MATTERS

TRANSFORMER-BASED LLMs



Neural network architecture, designed to learn patterns in language at massive scale

Convert language into numeric representations

Narrative framing (Van Hulst et al., 2025)

Frame Codebook

1. Moral judgment
2. Threat appraisal
3. Identity signaling
4. Care/harm concerns

Moral Judgment

Conceptual Definition: This frame interprets events through moral evaluation, emphasizing wrongdoing, injustice, blame, and accountability.

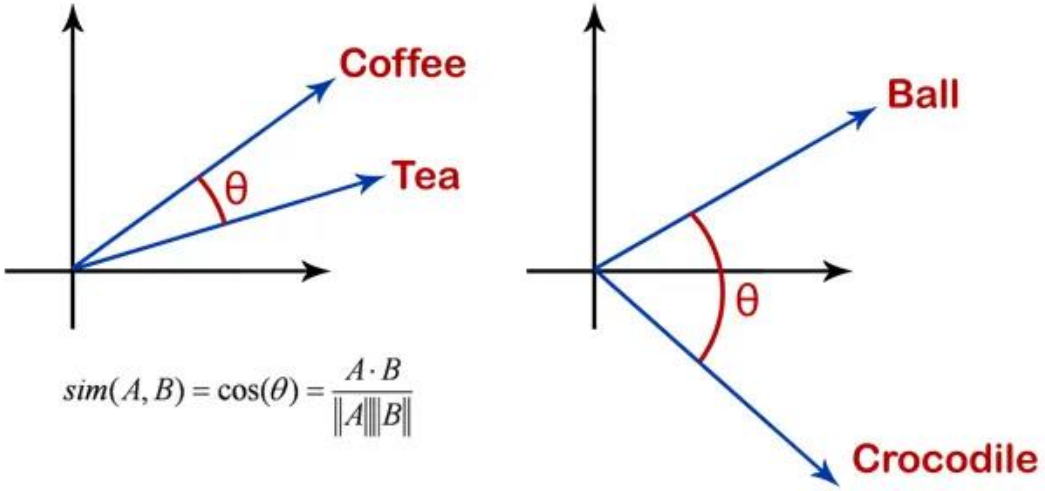
Core Interpretive Elements: attribution of responsibility, moral condemnation (good vs evil), calls for punishment or accountability, violations of norms or international law, moral outrage

Operational Description (Embedding Anchor Text): Language that frames events as morally wrong or unjust, assigns blame or responsibility, emphasizes violations of norms or law, and calls for accountability, punishment, or condemnation.

```r
frames <- tibble::tribble(
  ~frame, ~description,
  "moral_judgment",      "Language that frames events as morally wrong or
                          unjust, assigns blame or responsibility, emphasizes
                          violations of norms or law, and calls for
                          accountability, punishment, or condemnation",
  "threat_appraisal",    "Language that emphasizes danger, escalation, or
                          catastrophic risk, highlighting threats to safety,
                          stability, or survival and projecting potential
                          future harm.",
  "identity_signaling",  "Language that signals group alignment or coalition
                          membership, emphasizes shared values or collective
                          identity, and distinguishes between ingroups and
                          outgroups.",
  "care_harm_concerns",  "Language that emphasizes suffering, humanitarian
                          concern, empathy, or the need to protect and assist
                          vulnerable individuals or communities."
)
```

All-MiniLM-L6-v2

Hugging Face



$$sim(A, B) = \cos(\theta) = \frac{A \cdot B}{\|A\|\|B\|}$$

1

Compute similarity between each tweet and each frame

2

Find the best matching frame for each tweet.

3

If best similarity is below threshold, label as "unclassified"

cleanedtweet	frame	sim
horrendous acts which need to be tracked to the people responsible, and bring them to account for their war crimes	moral_judgment	.81
ukrainian refugees need our help to start their new lives.	care_harm_concerns	.89

```
assign_frames <- function(
  emb_block,
  frame_emb,
  frame_labels,
  min_sim = 0.20
```

SEMANTICALLY DISTINCT FRAMES: VALIDATE

Vague

Language about suffering, outrage, atrocities, and the war.

Better

Language that morally condemns wrongdoing by an actor, assigns blame or responsibility, describes violations of norms or international law, and calls for accountability, punishment, or justice

More – valenced language, less + language during early stages

1. Moral judgment
2. Threat appraisal***
3. Identity signaling
4. Care/harm concerns

1

LLMs expand what we can measure without replacing theory

2

LLMs make naturalistic, large-scale social psychological processes observable

3

Traditional methods
+
LLMs

Acknowledgments

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Thank you!

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